DONGSHENG ZUO

Postdoctoral Fellow ◇ Department of Computer Science & Engineering
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RESEARCH INTERESTS

- Machine learning for electronic design automation
- LLM agents for chip design automation
- Emerging architectures for efficient large language model inference

EDUCATION

The Hong Kong University of Science and Technology Sep. 2022 – June. 2026
Ph.D. Microelectronics

The Hong Kong University of Science and Technology Sep. 2021 – Jul. 2022
M.S. Electronic Engineering

Harbin Institute of Technology Sep. 2017 – Jul. 2021
B.Eng. Microelectronics
B.Eng. Data Science

EXPERIENCE

Postdoctoral Fellow June 2026 – Present
The Chinese University of Hong Kong
Hong Kong SAR

Research Intern Jul. 2023 – Dec. 2023
Huawei Noah's Ark Lab
Shenzhen, China

VLSI Physical Design Intern Mar. 2022 – Jun. 2022
NVIDIA Corporation
Beijing, China

Design Verification Intern Jun. 2021 – Aug. 2021
Cambricon Technologies Co., Ltd.
Beijing, China

FPGA Algorithm Intern Jan. 2021 – Apr. 2021
Tensorchip Technologies Co., Ltd.
Beijing, China

Research Assistant Jun. 2020 – Aug. 2020
University of Florida
Remote

AWARDS AND HONORS

[A3] **Best Paper Award Nomination**,
Asia and South Pacific Design Automation Conference (ASP-DAC), 2025.

[A2] **Best Paper Award**,
ACM/IEEE International Workshop on Machine Learning for CAD (MLCAD), 2023.

[A1] **Second Prize (National Finals) & Empyrean Special Award**,
National Undergraduate Embedded Chip and System Design Contest – Intelligent Recognition Chip Design Track, 2021.

PUBLICATIONS

First-Authored

[F6] **Dongsheng Zuo**, Rui Qiu, Qiang Xu, "GA^2-CT: GPU-Accelerated Genetic Algorithm for Low-Power Compression-Tree Rewiring", Asia and South Pacific Design Automation Conference (**ASP-DAC**), Tokyo, Japan, Jan. 2027.

[F5] **Dongsheng Zuo**, Jiadong Zhu, Yang Luo, Yuzhe Ma, "PrefixAgent: An LLM-Powered Design Framework for Efficient Prefix Adder Optimization", ACM Transactions on Design Automation of Electronic Systems (**TODAES**), 2026.

[F4] **Dongsheng Zuo**, Yang Luo, Jiadong Zhu, Yuzhe Ma, "TAP-MAC: Low-Power Optimization of Multipliers and Multiply-Accumulators via Topology-Aware Power Modeling", IEEE/ACM International Conference on Computer-Aided Design (**ICCAD**), San Jose, Nov. 2026.

[F3] **Dongsheng Zuo**, Jiadong Zhu, Yikang Ouyang, Yuzhe Ma "RL-MUL 2.0: Multiplier Design Optimization with Parallel Deep Reinforcement Learning and Space Reduction", ACM Transactions on Design Automation of Electronic Systems (**TODAES**), 2025.

[F2] **Dongsheng Zuo**, Jiadong Zhu, Chenglin Li, Yuzhe Ma, "UFO-MAC: A Unified Framework for Optimization of High-Performance Multipliers and Multiply-Accumulators", ACM/IEEE International Conference on Computer-Aided Design (**ICCAD**), New Jersey, Oct. 27-31, 2024.

[F1] **Dongsheng Zuo**, Yikang Ouyang, Yuzhe Ma, "RL-MUL: Multiplier Design Optimization with Deep Reinforcement Learning", ACM/IEEE Design Automation Conference (**DAC**), San Francisco, Jul. 09-13, 2023.

Co-Authored

[C12] Tairan Cheng, Yi Liu, **Dongsheng Zuo**, Zhengyuan Shi, Hongji Zhang, Xiangfei Hu, Maoshuo He, Hao Yan, Qiang Xu, "HINT: Toward an Executable Hardware-Intent Intermediate Representation Layer for LLM-Driven RTL Generation", Asia and South Pacific Design Automation Conference (**ASP-DAC**), Tokyo, Japan, Jan. 2027.

[C11] Weihao Wang, **Dongsheng Zuo**, Xiaoxiao Liang, Hongyuan Liu, Yuzhe Ma, "RT-DRC: Accelerating Design Rule Checking Using GPU Ray-Tracing Cores", Asia and South Pacific Design Automation Conference (**ASP-DAC**), Tokyo, Japan, Jan. 2027.

[C10] Kexin Chen, Yi Liu, **Dongsheng Zuo**, Xuhang Wang, Qiang Xu, Gang Chen, "ROOT: Trajectory-Guided RTL Optimization via Imitation Learning from Synthesis-Selected Rewriting", IEEE International Conference on Computer Design (**ICCD**), Hong Kong, Nov. 2026.

[C9] Yikang Ouyang, Yuchao Wu, **Dongsheng Zuo**, Subhendu Roy, Tinghuan Chen, Zhiyao Xie, and Yuzhe Ma, "SMART-GPO 2.0: Gate-Level Enhanced Sensitivity Measurement with Fast and Accurate Estimation for Glitch Power Optimization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), Early Access, Oct. 2026.

[C8] Yikang Ouyang, Yang Luo, **Dongsheng Zuo**, Yuzhe Ma, "Retrieve, Schedule, Reflect: LLM Agents for Chip QoR Optimization", ACM/IEEE International Symposium on Machine Learning for CAD (**MLCAD**), Jeju, Sep. 2026.

[C7] Rui Qiu, Guoyun Gao, Zhiyuan Du, Ni Yang, Zefan Li, **Dongsheng Zuo**, Yunwei Tong, Peiyi He, Hangming Zhang, Hao Jiang, Lain-Jong Li, Han Wang, Can Li, "Monolithic 3D Integration of MoS_2 eDRAM and RRAM for Analog In-Memory Attention Computing", IEEE International Electron Devices Meeting (**IEDM**), San Francisco, Dec. 2025.

[C6] Zhihai Wang, Jie Wang, Xilin Xia, **Dongsheng Zuo**, Lei Chen, Yuzhe Ma, Jianye HAO, Mingxuan Yuan, Feng Wu "Computing Circuits Optimization via Model-Based Circuit Genetic Evolution", International Conference on Learning Representations (**ICLR**), Singapore, Apr. 2025.

[C5] Jiadong Zhu, **Dongsheng Zuo**, Yuzhe Ma, "A Holistic FPGA Architecture Exploration Framework for Deep Learning Acceleration", Asia and South Pacific Design Automation Conference (**ASP-DAC**), Tokyo, Japan, Jan. 2025.

[C4] Yikang Ouyang, Yuchao Wu, **Dongsheng Zuo**, Subhendu Roy, Tinghuan Chen, Zhiyao Xie, and Yuzhe Ma, "SMART-GPO: Gate-Level Sensitivity Measurement with Accurate Estimation for Glitch Power Optimization", Asia and South Pacific Design Automation Conference (**ASP-DAC**), Tokyo, Japan, Jan. 2025. **(Best Paper Award Nomination)**

[C3] Zhihai Wang, Jie Wang, **Dongsheng Zuo**, Ji Yunjie, Xilin Xia, Yuzhe Ma, Jianye Hao, Mingxuan Yuan, Yongdong Zhang, Feng Wu, "A Hierarchical Adaptive Multi-Task Reinforcement Learning Framework for Multiplier Circuit Design", International Conference on Machine Learning (**ICML**), Vienna, Austria, Jul. 2024.

[C2] Chen Chen, Guangyu Hu, **Dongsheng Zuo**, Cunxi Yu, Yuzhe Ma, Hongce Zhang, "E-Syn: E-Graph Rewriting with Technology-Aware Cost Functions for Logic Synthesis", ACM/IEEE Design Automation Conference (**DAC**), San Francisco, Jun. 2024.

[C1] Yikang Ouyang, Sicheng Li, **Dongsheng Zuo**, Hanwei Fan, Yuzhe Ma, "ASAP: Accurate Synthesis Analysis and Prediction with Multi-task Learning", ACM/IEEE Workshop on Machine Learning for CAD (**MLCAD**), Utah, Sep. 2023. **(Best Paper Award)**